NUMERICAL METHODS FOR PDES
Lecture 3

Q 3.2

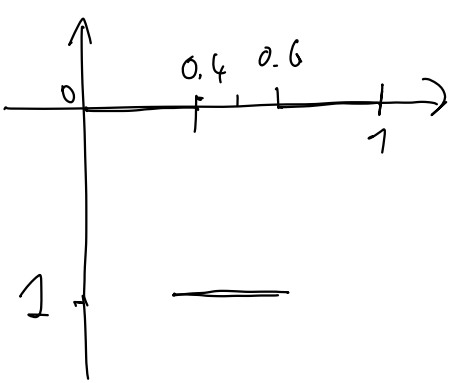
Model problem (strong form). Find $u \in C^2(0,1)$:

$$\begin{cases} -u'' = f & \Omega = (0,1) \\ u(0) = 0 = u(1) \end{cases}$$

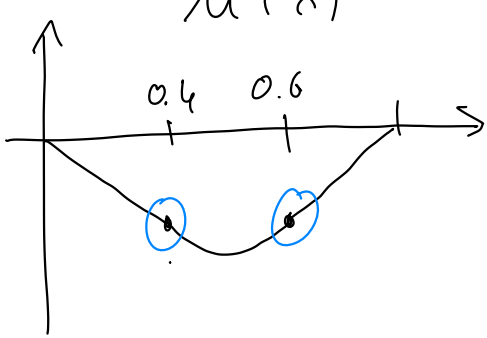
Examples

①

$$f(x) = -\chi_{[0.4,0.6]}(x)$$



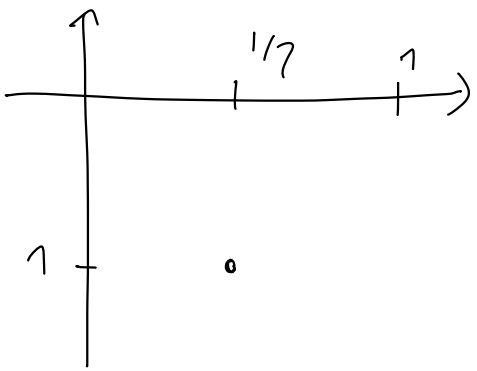
$u(x)$

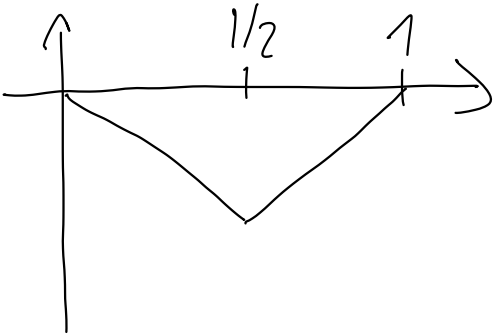


$u \in C^1([0,1])$

(2) $= \begin{cases} -1 & \text{if } x = 1/2 \\ 0 & \text{otherwise} \end{cases}$

$f(x) = -\delta_{1/2}(x) \Rightarrow u \in C^0([0,1])$





Dirichlet principle: elastic string minimizes the potential energy:

$$J(v) = \frac{1}{2}\int_0^1 (v')^2 dx - \int_0^1 f v \, dx$$

subject to $v(0) = v(1) = 0$

Calculus of variations: If u is minimizer of J, then

the "first variation" must be zero:

$$0 = \frac{d}{d\varepsilon} J(u+\varepsilon v)\Big|_{\varepsilon=0}$$

$$= \frac{d}{d\varepsilon}\left(\frac{1}{2}\int_0^1 (u+\varepsilon v)'(u+\varepsilon v)'\,dx - \int_0^1 f(u+\varepsilon v)\,dx\right)\Big|_{\varepsilon=0}$$

$$= \frac{d}{d\varepsilon}\left(\frac{1}{2}\int_0^1 (u'+\varepsilon v')(u'+\varepsilon v')\,dx - \int_0^1 f(u+\varepsilon v)\,dx\right)\Big|_{\varepsilon=0}$$

$$= \left(\int_0^1 (u'v' + \varepsilon v'v')\,dx - \int_0^1 f v\,dx\right)\Big|_{\varepsilon=0}$$

$$= \int_0^1 u'v'\,dx - \int_0^1 f v\,dx$$

$$\Rightarrow \boxed{\forall v \in (?)\ \exists \int_0^1 u'v'\,dx = \int_0^1 f v\,dx}$$

variational or weak formulation

v is a test function

To fix the test space, consider the PDE

$$-u'' = f$$

and "test" with v such that $v(0) = 0 = v(1)$ and multiply the PDE:

$$-u''v = fv$$

and integrate:

$$-\int_0^1 u''v\,dx = \int_0^1 fv\,dx.$$

Integrating by parts the right-hand side,

$$\int_0^1 u'v'\,dx - u'v\Big|_{x=0}^{x=1} = \int_0^1 fv\,dx$$

$u'v\big|_{x=0}^{x=1} = 0$ due to the homog. b.c.

$$\Rightarrow \int_0^1 u'v'\,dx = \int_0^1 fv\,dx \qquad (*)$$

Principle of virtual work

Fixing the test space V

Wrong guess: $V = \{v \in C^1([0,1]) : v(0) = 0 = v(1)\}$

Issues:

(1) $f = \delta_{1/2}$ seen that $u \notin C^1([0,1])$

(2) even if $f \in C^0([0,1])$

$\exists!$ of u satisfying $(*)$ is not guaranteed

The problem is that such V

is <u>not</u> complete with respect to the (natural) scalar product

$$(u,v)_1 := \int_0^1 u' v' \, dx$$

$$\left(\Rightarrow \text{(semi) norm} \quad |v|_1 := (v,v)_1 \right)$$

The correct space should be complete...

Right guess: the H_0^1 Hilbert space.

Def: $L^2(0,1) = \left\{ v : (0,1) \to \mathbb{R} : \|v\|_2 := \left(\int_0^1 |v|^2 dx \right)^{1/2} < +\infty \right\}$

↑ space of square integrable functions

Hilbert with L^2-product $\quad (w,v)_0 := \int_0^1 w v \, dx \quad \forall w, v \in L^2(0,1)$

$$H^1(0,1) = \left\{ v \in L^2(0,1) : v' \in L^2(0,1) \right\}$$

Here, with v' we denote a generalised notion of derivative, the weak derivative, such that if $v \in C^1(0,1)$ then v' coincides with the classical derivative.

$$H_0^1(0,1) = \overline{C_0^\infty(0,1)}^{H^1(0,1)}$$

In 1D, H has continuous representatives (by Sobolev Embedding)

we can write $\Rightarrow$ $H_0^1(0,1) = \{v \in H^1(0,1) : v(0) = 0 = v(1)\}$

(in general by Trace theorem)

Theorem: $\left(H_0^1(\cdot,\cdot), (\cdot,\cdot)_1\right)$ is a Hilbert space

- H^1 functions are not necessarily C^1
- H^1 is the closure of C^1 w.r.t. $(\cdot,\cdot)_1$

Weak formulation

Let $f \in V' =$ dual space of H^1

(WP) Find $u \in V := H_0^1(0,1)$:

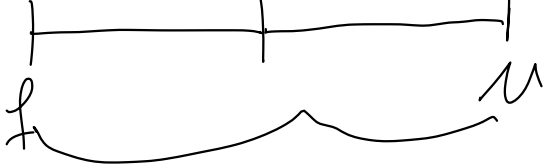
$$\int_0^1 u' v' \, dx = \langle f, v \rangle = \forall v \in V$$

(duality pairing)

(if $f \in L^2(0,1)$ then $= \int_0^1 f v \, dx$)

but instead, $\delta_{1/2} \notin L^2$. $\langle \delta_{1/2}, v \rangle = v(1/2)$ $= (f, v)_0$

$V' = (H_0^1)'$ $\quad L^2 \quad$ $H_0^1 = V$



(roughly, we have regularity shift by 2)

Theorem: (VP) is well posed for all $f \in V'$.

Proof: (later) based Lax-Milgram Lemma (or identification with minimization problem)

stability (by energy method)

$$= |u|_1^2 = \int_0^1 u' u' = \langle f, u \rangle$$

$$= (u, u)_1$$

$$\leq \|f\|_{V'} \|u\|_V$$

(Schwartz)

$$\Rightarrow \boxed{|u|_1 \leq \|f\|_{V'}}$$

continuous dependence on the data